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An agility test to measure performance in Para ice hockey

Test presentation and test-retest reliability

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***Abstract*— Para ice hockey is a Paralympic discipline in which athletes with a physical impairment at the lower part of the body play sitting on a sledge. To take part in the competitions, athletes have to meet the minimum disability criteria. In recent years, the International Paralympic Committee has promoted the necessity to move towards evidence-based assessment of the impact of the impairment on performances. In Para ice hockey, performance is related to the ability of making sprint and to the maneuverability in curves and in handling the puck. A new agility test to measure athletes' performance in terms of taking curves is here proposed and assess for its intraday reliability. Two acquisitions were done in order to collect respectively left and right curves. Six athletes performed three repetitions for each acquisition (left and right), for a total of six trials. The time required to complete the test (execution time) was collected for each trial. In addition, blade angle and trunk angle were calculated. Blade angle is the inclination of the blade with respect to the horizontal; whereas trunk angle is the inclination of the trunk with respect to the horizontal. The intraclass correlation coefficient, the absolute and percent standard error of measurement, and the mean difference were calculated between the three repetitions for both left and right side, for all variables. For kinematic and temporal variables, results showed high value of intraclass correlation coefficient (>0.82), low value of standard error of measurement (<4.7%), and low value of mean difference (<1.75° and 0.18s) for both left and right curves. A difference ($p<0.01$) was found between left and right curves for the kinematic variables; whereas execution time did not show any difference between sides. Finally, the first curve was different ($p<0.01$) compared to the others in both blade angle and trunk angle for left and right curves. In conclusion, because of the low within-subject variation, this agility test may be considered sport-specific measures of performance for evidence-based classification. In addition, it can be used by coaches to make personalized training.**

Keywords—Paralympic sport; Classification; Performance; Sport specific; Biomechanics.

I. INTRODUCTION

In 2018 the XII Paralympic Winter Games are taking place in PyeongChang (South Korea). The increasing in attention towards the Paralympic sports has been demonstrated by the augmented broadcasting and media personnel [1]. A greater interest is also shown by the ever more scientific researches on the biomechanics, physiology and equipment of Paralympic Winter sports [2],[3],[4],[5],[6]. Among Winter sports, alpine skiing and Para ice hockey are the two with the highest increase in athletes' participation in the last Paralympic Winter Games [7].

Para ice hockey is a team sport played on ice by two teams of six players. It is a Paralympic discipline, which follows the rules of the International Ice Hockey Federation. It is played in a rink of approximately 60 x 30 m and the game consists of three 15-minutes periods [8]. Two are the major differences between Para ice hockey and ice hockey. The first difference is in equipment since Para hockey athletes compete seated and strapped on a sledge. The sledge is composed of a seat mounted on a metallic frame with a pair of blades under it. In addition, athletes are equipped with a couple of modified hockey sticks, which are used to generate propulsion, maneuver the sledge (agility), and handle the puck. The second great difference with respect to ice hockey is the penalty of teeing. The teeing is the action of hitting another player by the front of the sledge. Teeing, obviously, does not exist in ice hockey.

The International Paralympic Committee aims to spread the sport among people with disabilities and to ensure fair and equal competitions. In order to achieve this purpose, impact of impairment should be minimized on performance outcome. Therefore, most Paralympic sports have minimum disability criteria to define athletes eligibility and a classification process to divide athletes in groups according to the impact of the impairment [9]. However, in some sport such as Para ice hockey, there is only one sport class of athletes who meet minimum disability criteria in one of the seven recognized physical impairment (limb deficiency, muscle power impairment, impaired range of motion, ataxia, athetosis, hypertonia, leg length difference) [10]. As a matter of fact, usually Para ice hockey players have a physical impairment at the lower part of the body, which simply does not allow them to play standing [10] (Fig. 1).

It has been shown that physical impairment influence game efficiency in many team sports such as wheelchair basketball [11] and wheelchair rugby [11]. It seems that Para ice hockey players with different impairments (double

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Fig. 1. Para ice hockey players compete sitting because of physical impairment at the lower limb does not allow them to play standing.

amputee, single amputee, spinal cord injury, and others) had no differences in game parameters such as shots resulting in a goal, good passes, good received passes, and efficiency of shots, passes, and receiving [12]. However, this lack of difference could be due to the fact that most of the teams included only players who had minimal functional disabilities, such as those with high residual body core control [12]. Physical impairment seems to influence also performance in terms of average and peak power while poling on the ergometer and in terms of time decrement in sprint test on ice [13]. Thefore, in order to spread the sport among people with disabilities, players with moderate to high impact of impairment should have the possibility to be well represented and competitive in this sport.

In last years, the International Paralympic Committee highlighted the necessity to move towards evidence-based assessment of the impact of the imparment on performances. This new system is composed of three steps [14]: development of valid measurements of physical impairments, development of standardized and sport-specific measurements of performance, and assessment of the relative association strength between the two. Due to the many aspects that influence performance of wheelchair team sports, there is no a single evidence-based test to assess performance level [15]. However, sprint and maneuverability were identified as good determinants of wheelchair sports performance [16]. Similarly to wheelchair sports, performance in Para ice hockey can be related to print and maneuverability (agility in taking curves and puck handling) [17]. The purpose of this study was to present a new measurement of performance in terms of maneuverability, specific for Para ice hockey. In addition, its intraday reliability was assessed in Para ice hockey athletes with different impairments and experience level. In order to achieve these purposes, an agility test composed of left and right curves was performed on ice, while the athlete was sitting on his personal sledge. To assess test-retest intraday reliability, the agility test is repeated three times for each side. Future effects of the study will be its integration in evidence-based assessment of the impact of the imparment on performances.

II. Methods

Six male para ice hockey players (age 33±8 years, weight 77±12 kg, height 180±9 cm, representing different impairment: above knee amputation, hemimelia, incomplete paraplegia, Phocomelia) participated as volunteer in this study. Athletes played in the Sportdipiù Tori Seduti team of Torino (Italy) and four of them played in the Italian National team. All athletes were well trained and right handed. Athletes were completely informed about the nature of the study. All of them signed an informed consent and were free to interrupt the tests at any moment. The research methods and protocols were standard and measurements were performed in accordance with the declaration of Helsinki.

A. Overall design and protocol

An agility test was conducted in order to test athlete's agility in performing curves while skating on ice at high speed. For all athletes, the tests were conducted one month later the beginning of the training season in order to allow complete training status recovery after summer rest period. The test set up was composed of two rows of disc cones placed with a distance of 2 m in between (Fig. 2). Each row was composed of four disc cones, with a space among following cones equal to 3 m. The disc cones of the two rows were shifted of a distance of 1.5 m. A camera (GoPro HERO 3+ Silver, resolution of 1280x720 and sample frequency of 120 Hz) was aligned with the row of red disc cones (Fig. 2), 3 m away from the first disc cone (defined according to the movement direction). The camera was fixed on a tripod 50 cm height. For the test, athletes were seated on their personal sledge, strapped as for competitive events, and both sticks were allowed. An adhesive marker was fixed on athlete back, in correspondence of the middle point between the two shoulders and a second marker was fixed on the middle point of the sledge seat backrest.

Before the beginning of the tests, 10-15 minutes were given to the athletes for a free warm-up. For the agility test two different acquisitions were done with the camera aligned to the row of red disc cones to collect respectively turning to the left (Fig. 2A) and to the right (Fig. 2B). Each acquisition was repeated 3 times with 30 s in between. A total of 6 trials were collected in the following order for all athletes: 3 trials for the left curves (Fig. 2A) and 3 trials for the right curves (Fig. 2B). In each trial, a total of 4 left (Fig. 2A) or 4 right (Fig. 2B) curves were recorded. The athlete was requested to approach and to perform the agility path from the starting point (marked as "S" in Fig. 2) and to perform a slalom among the disc cones in the shortest time possible. If one of the cones was touched by the sledge blades or by the sticks, the trial was repeated.

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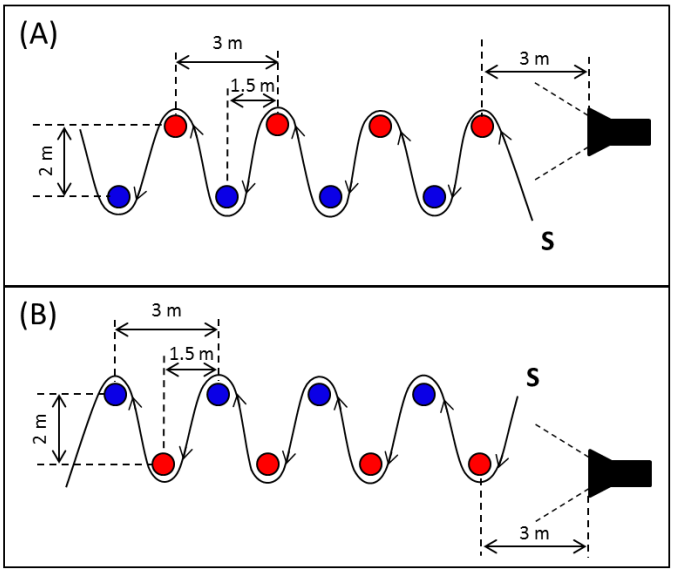


Fig. 2. Two acquisitions to collect: (A) left curves and (B) right curves. A GoPro camera was placed in correspondance with the line of red disc cones, 3 m away from the first cone (from the starting position "S").

B. Data analysis

For all the curves of each of the six trials (three for the left and three for the right) the following kinematic and temporal variables were calculated: blades angle, trunk angle, and execution time. Blade angle was calculated as the angle between the blade line and the horizontal line (Fig. 3A). The blade line was drawn between the contact point of the blade with the ice and the marker fixed at the middle point of the sledge seat backrest. Trunk angle was calculated as the angle between the trunk line and the horizontal line (Fig. 3B). The trunk line was drawn between contact point of the blades with the ice and the marker fixed on the back in the middle point of the shoulders. Execution time (Fig. 4) was measured from the first curve (when the blades were in correspondence of the disc cones row) to the instant in which the front of the sledge arrived in correspondence of the last cone (on the opposite disc cones row). The kinematic and the temporal variables were assessed by video analysis using Kinovea software, by an expert operator in order to reduce measurement errors. Kinovea allows calculating both angle and time, errors related to the temporal variable could be 8 to 16 milliseconds as the start and end instants could be identified one frame earlier or later. It was expected that athletes with the ability to voluntary control the core muscles make curves mainly tilting the pelvis, the shoulder or a combination of both, depending for example on their skating speed or curve radius. Therefore, the blade angle was calculated because related to the sledge inclination during the curves; whereas the trunk angle is related to the inclination of both sledge and athlete's trunk. Both the blade angle and the trunk angle were calculated for each of the four curves when the athlete was in correspondence of the disc cone and his back in line with the camera.

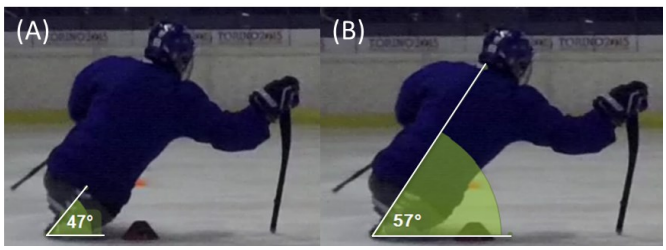


Fig. 3 Kinematic variables evaluated: (A) blade angle and (B) trunk angle.

C. Statistical analysis

In order to assess test-retest intraday reliability of the agility test, intraclass correlation coefficient (ICC), standard error of measurement absolute (SEM) and percent (SEM%), and mean difference (MD) were calculated among the three repetitions (trials) for all kinematic and temporal variables [18].

A Wilcoxon test was applied to the kinematic and execution time variables to assess statistical differences between the left and the right curves.
A Friedman test was used to assess the statistical difference among the four curves for the kinematic variables. Tukey post hoc was used when necessary. The statistical analysis was done using SPSS software and the statistical significance was set as $p<0.05$.

III. Results

During the test, none of the athletes touched the cones; therefore only three repetitions were performed by each

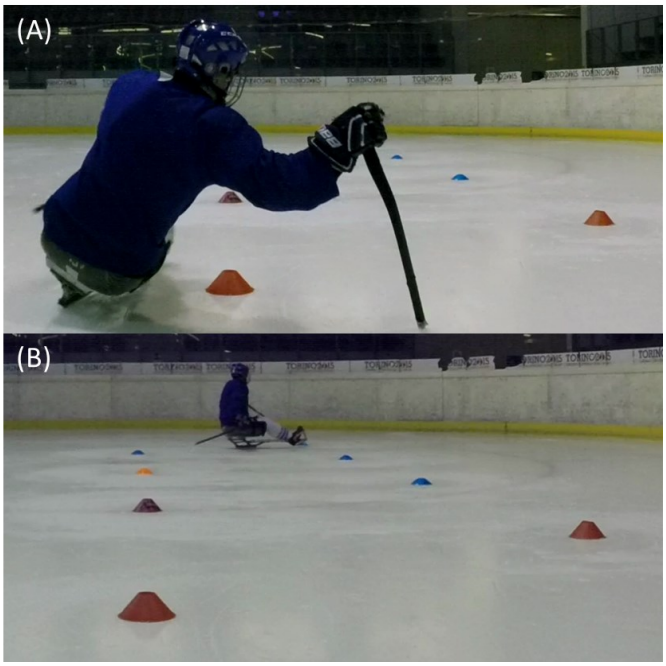


Fig. 4 The execution time was measured between two instants: (A) when the blades were parallel to the disc cones row and (B) when the front of the sledge arrived in correspondence of the last cone (on the opposite row).

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player. For the blade angle, trunk angle, and execution time, mean values and standard deviations among the four curves for each of the three left and the three right trials are reported in Table I. In addition, overall mean value and standard deviation for all trials are reported in the last column of Table I for both left and right curves. In Table II are reported intraday test-retest results (ICC, SEM, SEM%, and MD) for the blade angle, trunk angle, and execution time in both left and right curves.

Comparing the left and right curves, the kinematic variables showed a statistical difference ($p<0.01$); whereas no statistical difference was found for the execution time (last column of Table I). Concerning the curves comparison, both sledge angle and trunk angle showed a statistical difference ($p<0.01$) between the first and the second curve on the left side; whereas for the right curves the difference was between the first and other three curves ($p<0.001$) for both kinematic variables.

IV. Discussion

The aim of this study was twofold: 1) to introduce a new measurement of Para ice hockey performance in terms of maneuverability and 2) to evaluate its test-retest intraday reliability including athletes with different level of impairment and experience level.

TABLE I. The mean and standard deviation for kinematic and temporal variables in each of the three trials and overall for all trials. (*) Statistical difference with the right sides.

Variable	***Curve side***	**Trials**			
		1	***2***	***3***	***Total***
Blade angle (°)	Left	57.5 (7.5)	58.5 (6.6)	57.4 (7.0)	57.8 (7.0) *
	Right	52.0 (7.7)	49.4 (7.1)	51.7 (8.1)	51.0 (7.6)
Trunk angle (°)	Left	61.3 (6.1)	61.7 (6.6)	61.0 (7.0)	61.3 (6.6) *
	Right	58.5 (5.2)	58.8 (4.9)	60.3 (5.2)	59.2 (5.1)
Execution time (s)	Left	11.3 (1.8)	11.6 (2.1)	11.6 (2.4)	11.5 (2.1)
	Right	11.4 (2.1)	11.6 (2.1)	11.5 (2.1)	11.5 (2.1)

TABLE II. Test-retest intraday reliability results for kinematic and temporal variables. The reliability is calculated by means of: the Intraclass correlation coefficient (ICC), the standard error of measurement (SEM), and the mean difference (MD).

Variable	***Curve side***	**Coefficient**		
		ICC	***SEM (SEM%)***	***MD***
Blade angle	Left	0.86	2.6° (4.6%)	0.72°
	Right	0.90	2.4° (4.7%)	1.75°
Trunk angle	Left	0.94	1.6° (2.6%)	0.47°
	Right	0.82	2.2° (3.6%)	1.22°
Execution time	Left	0.98	0.30 s (2.6%)	0.18 s
	Right	0.99	0.20 s (1.8 %)	0.08 s

It has been shown that in team sports, such as Para ice hockey, ability to perform maximal, short, and repeated sprint (repeated-sprint ability) is fundamental in increasing power and, thereby, performance [18]. For this reason, in this sport discipline sprint performance has been wildly investigated in relation to the strength and power generated in upper-body exercises [19],[20] and to the physiological demands [4]. However, a second important determinant of performance exists. It is almost mandatory for Para ice hockey players have a good maneuvering agility when handling the puck, facing an opponent, and taking curves. In particular, it is fundamental the ability of the players to perform sprints in those technically complex situations [17].

To assess players' agility, a modified version of ice-hockey agility test was already proposed for Para ice hockey players and assessed for its reliability [21]. The proposed test consisted of multiple turns, which start and end at the same point, and have to be completed in the shortest time as possible, while handling the puck. The execution time was recorded by a pair of light sensors fixed at the start and stop line. A good reliability was shown for performance in terms of execution time: intraclass correlation coefficient equal to 0.94, standard error of measurement equal to 1.14 s, standard error of measurement percent equal to 4.3% [21]. It was also shown that this agility correlate well with laboratory aerobic and anaerobic capacities, showing a good association between ability to sprint in technically complex situations and maximal strength and power [17]. However, nothing was known about performance in terms of athletes' ability to take curves during this agility test. It can be supposed that athletes with low level of impairment and better performance are the same who have the greater sledge maneuverability and ability to keep the balance while tilting, as it happens for example in wheelchair rugby [22]. Therefore in this study an agility test was proposed in order to assess performance in both execution time and kinematics.

The proposed agility test had a different design compared to the one proposed in [21]: different cones position, different ending point, and it was performed without the puck. This decision was taken since the aim of the current study was to assess maneuverability in terms of taking curves and not in puck handling. Therefore, the agility circuit was designed together with athletes and the coach in order to reproduce curves as much similar as the ones usually performed during a match. In the present study to assess athletes' ability to make curves, a kinematic analysis was used. Kinematic analysis is usually evaluated using wearable inertial sensors [23], however since the jersey and protective gear wore by athletes this collecting system could not be used. In addition, due to the challenging environment conditions, collecting video using standard cameras system would be technically demanding. Therefore, a single GoPro camera with high frame acquisition rate was used.

Considering the execution time, very high intraclass correlation coefficient, low absolute and percent standard error of measurement, and low mean difference suggest high test-retest intraday reliability for the propose agility test. Compared to the previous version of this test [21], this new agility test showed higher intraclass correlation coefficient and

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lower standard error of measurement percent. Despite the different impairments and experience levels, the low within-subject variation suggests that this test could be considered to assess athletes' agility performance. Considering the kinematics results, a very high intraclass correlation coefficient was found both for blade angle and trunk angle. The standard error of measurement percent was slightly higher for blade angle than for trunk angle, suggesting the latter as more suitable variable to evaluate athletes kinematic. However, overall standard error of measurement percent was small for both kinematic variables even if athletes with different experience levels and impairments were included in the study.

Comparing the left and right curves, no statistical difference were found in execution time between left and right side (Table I). This was expected since the two set up were extremely similar. The comparable time of execution between the two designs may also suggest that it could not be different for athletes accelerating from standstill in a left of right curve. In contrast, a statistical difference was found in blade angle and trunk angle between left and right curves (Table I). In particular, smaller blade angle and trunk angle were found for right curves; this means that athletes tilt more on this side compared to the left. This result could be influence by their dominant side (right for all athletes), which might affects more one side compared to the other.

In curves comparison, a statistical difference was found between the first curve and the others in both left and right side for blade angle and trunk angle. This difference may be due to the fact that athletes approach the first curve starting from a standstill condition, whereas they were already in motion for the subsequent three curves. Therefore, for future studies it should be considered to discard the first curve from the analysis, since it may be alter the results.

V. Limitations

The small number of athletes who participated may be a limitation of this study. However, in future study to assess the performance in terms of agility and maneuverability athletes from other teams can be involved. Including a higher number of athletes allows evaluating also the impact of different impairments (amputations, paraplegia, and others) and, possibly, the impairment side on the maneuverability results. A second limitation of this study can be that during the test athletes were equipped with their jersey and protective gear. This may affect the position of the marker, which might move with respect to the middle point between the two shoulders and therefore the accuracy of the trunk angle.

VI. Conclusion

The present study aims to introduce a new agility test as sport-specific measure of performance in terms of maneuverability. Moreover, the test-retest intraday reliability was assessed including Para ice hockey athletes with different physical impairments and experience level. From the results it is possible to point out that, despite the variability in athletes experience and impairment level, the agility test showed a low within-subject variation both in execution time and kinematics. These findings allow supposing that this test may be included in evidence-based classification process as sport-specific measures of performance.

In addition, the difference between left and right side may suggest that this test can identify the athlete's dominant side. This information may be useful for coaches to make personalized training in order to potentiate athlete's weaker side.

There are still open points, which can be considered for future studies in Para ice hockey. The first one is how much personalized equipment can influence athletes' performance. To date, sledges design is very similar for all athletes, independently from their impairment. It can be supposed that sledge seat with higher backrest may help athletes with high impact of impairment (spinal cord injuries) in keeping the balance while skating. In order to make personalized equipment, new possibilities are offered by manufacturing technique, such as the additive ones [24]. The second open point is the use of simulations [25]. Simulations can be used to better understand the effect of different impairments on performance, but can also be used together with the manufacturing technique to determine and design the most suitable sledge for each athlete.

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